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CONVEX n-OMINOES

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Abstract. Unit squares having their vertices at integer points in the Cartesian plane are called *cells*. A connected union of n distinct cells having no finite cut set is an *n-omino*. Two n-ominoes are the same if one is mapped onto the other by a translation of the plane. An n-omino is *convex* if the cells in each row and each column form a connected strip. When viewed from a distance, most convex n-ominoes resemble rods tilted 45° from the vertical with horizontal (and vertical) thickness roughly equal to 2.37597. If $c(n)$ denotes the number of convex n-ominoes, then $c(n) \sim f\gamma^n$, where $\gamma = 2.30914$ and $f = 2.67564$. (It is understood that all constants are accurate to within $\pm\frac{1}{2}$ in the last place.)

1. Introduction

Unit squares having their vertices at integer points in the Cartesian plane are called *cells*. A connected union of n distinct cells having no finite cut set is an *n-omino* (also called an *animal*). If two animals differ by a translation of the plane, they are considered to be the same. Klarner and Rivest [4] have defined a *convex* animal to be an animal such that every intersection with a vertical or horizontal line is a (possibly empty) interval. Let $c(n)$ denote the number of convex n-ominoes. Klarner and Rivest showed that $c(n)^{1/n} \sim \gamma$ and found an effective procedure for calculating γ.

We will study the typical shape of convex animals and obtain improved asymptotic results for $c(n)$. All numerical constants are understood to be accurate to within $\pm\frac{1}{2}$ in the last place.

I would like to thank David Klarner for discussing [4] with me and pointing out the derivation of (3).

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2 Two generating functions

We will use the method of [3] to develop multivariate generating functions for "trapezoids" and "parallelograms". These form the basis of our later analyses.

Both trapezoids and parallelograms are convex animals. Hence they can be specified by giving two finite sequences of integers $l_1, l_2, \ldots$ and $r_1, r_2, \ldots$ such that $l_i < r_i$ for all i; the animal consists of the set of cells with centers $(i + \frac{1}{2}, j + \frac{1}{2})$, where $l_i \leq j < r_i$. In these terms, a *trapezoid* is defined by a sequence such that $l_i \leq l_{i+1}$ and $r_i \geq r_{i+1}$. A *parallelogram* (to the right) is defined by a sequence such that $l_i \leq l_{i+1} < r_i < r_{i+1}$.

Let $T(x, y)$ be the generating function for trapezoids such that the number with n cells and base length b is the coefficient of $x^n y^b$. Apply [3, (17)] with

$$f(a, b) = \begin{cases} 0, & a < b, \\ (a - b + 1), & a \geq b, \end{cases}$$

(1)

$$g(a) = \begin{cases} 0, & a \leq 0, \\ 1, & a > 0, \end{cases}$$

to obtain

$$T(x, y) = \frac{xy}{1 - xy} + \frac{1}{2\pi i} \oint \frac{T(x, s)}{1 - xy/s} \frac{1}{(1 - xy)^2} \frac{ds}{s}$$

(2)

$$= \frac{xy}{1 - xy} + \frac{1}{(1 - xy)^2} T(x, xy),$$

since $T(x, s)/(s - xy)$ has a simple pole at $s = xy$. Iterating (2) leads to

$$T(x, y) = \sum_{k=1}^{\infty} \frac{x^k y(1 - x^k y)}{\prod_{n=1}^{k} (1 - x^n y)^2}. \tag{3}$$

Let $P(x, t, y_1, y_2)$ be the generating function for parallelograms such that the coefficient of $x^n t^h y_1^{b_1} y_2^{b_2}$ is the number of n-celled parallelograms of height h with lower and upper bases b_1 and b_2. Apply [3, (17)] with

$$f(a, b) = \min(a, b)t\,,$$

(4)

$$g(a) = \begin{cases} 0, & a \le 0, \\ t\,y_2^a, & a > 0, \end{cases}$$

to obtain

$$P(x, t, y_1, y_2) = \frac{txy_1y_2}{1 - xy_1y_2} + \frac{1}{2\pi i}\oint \frac{xy_1\,tP(x, t, s, y_2)}{(1 - xy_1)(s - 1)(s - xy_1)}\,ds$$

$$= \frac{txy_1y_2}{1 - xy_1y_2} + \frac{xy_1t}{1 - xy_1}\left(\frac{P(x, t, 1, y_2)}{1 - xy_1} - \frac{P(x, t, xy_1, y_2)}{1 - xy_1}\right)$$

Iterating we obtain

$$P(x, t, y_1, y_2) = P_1(x, t, y_1, y_2) + P_2(x, t, y_1)P(x, t, 1, y_2),$$

where

(5a) $$P_1(x, t, y_1, y_2) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}x^{k(k+1)/2}(ty_1)^k y_2}{(1 - x^k y_1 y_2)\prod_{n=1}^{k-1}(1 - x^n y_1)^2}\,,$$

(5b) $$P_2(x, t, y_1) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}x^{k(k+1)/2}(ty_1)^k}{\prod_{n=1}^{k}(1 - x^n y_1)^2}\,.$$

Hence $P(x, t, 1, y_2) = P_1(x, t, 1, y_2)/(1 - P_2(x, t, 1))$ and so

(6) $$P(x, t, y_1, y_2) = P_1(x, t, y_1, y_2) + \frac{P_1(x, t, 1, y_2)P_2(x, t, y_1)}{1 - P_2(x, t, 1)}\,.$$

This is a simple adaptation of the procedure Klarner and Rivest used to obtain $P(x, 1, y, 1)$.

3. Determination of γ

It was shown in [4] that $\gamma = r^{-1}$, where r is the radius of convergence of

$$P(x) = P(x, 1, 1, 1) = P_1(x, 1, 1, 1)/(1 - P_2(x, 1, 1))\,.$$

Since the numerator and denominator of $P(x)$ are analytic for $|x| < 1$, any singularities of $P(x)$ with $|x| < 1$ must be poles due to roots of $P_2(x, 1, 1) = 1$. We evaluated

$$\frac{1}{2\pi i} \oint \frac{P_2'(z, 1, 1)}{P_2(z, 1, 1) - 1} \, dz$$

numerically on the circle $|z| = \frac{1}{2}$ by various methods and obtained the value 1.00. Hence $P_2 - 1$ has exactly one zero inside the circle $|z| = \frac{1}{2}$ and it is simple. Since $P(x)$ has non-negative coefficients $p(n)$, it has a singularity at $x = r$ which, by the previous discussion, is a simple pole due to a simple root of $P_2(x, 1, 1) = 1$. Because the series (5b) for $P_2(x, 1, 1)$ is an alternating series with strictly decreasing terms when $x^k < (1 - x^k)$, upper and lower bounds can be found for $P_2(x, 1, 1)$ by truncating the series. (This idea is used to compute all the numerical constants.) Thus it is easy to use a computer to determine r. We found

$$r^{-1} = \gamma = 2.30913859330$$

by using the first ten terms in (5b). We also obtained $P_1(r, 1, 1, 1)/(rP_2'(r, 1, 1)) = 0.29745$. Thus

$$p(n) \sim 0.29745\gamma^n \tag{7a}$$

since the pole is simple. The total number of parallelograms is asymptotic to

$$2p(n) \sim 0.59491\,\gamma^n \,. \tag{7b}$$

4. The number of convex n-ominoes

Let the sequences $l_1, l_2, \ldots$ and $r_1, r_2, \ldots$ define a convex animal slanted to the northeast. Define I to be the maximum i such that $l_i > l_{i+1}$. Similarly, define J to be the minimum j such that $r_{j-1} > r_j$. The three parts of the convex animal are determined by those l_i and r_j with i in the ranges $i \leq I$, $I < i < J$, $i \geq J$ (see Fig. 1). Let $c^*(n)$ be the number of convex n-ominoes with $J > I + 1$, i.e., those animals with a non-empty parallelogram part slanting to the northeast. Since $T(x, 1)$

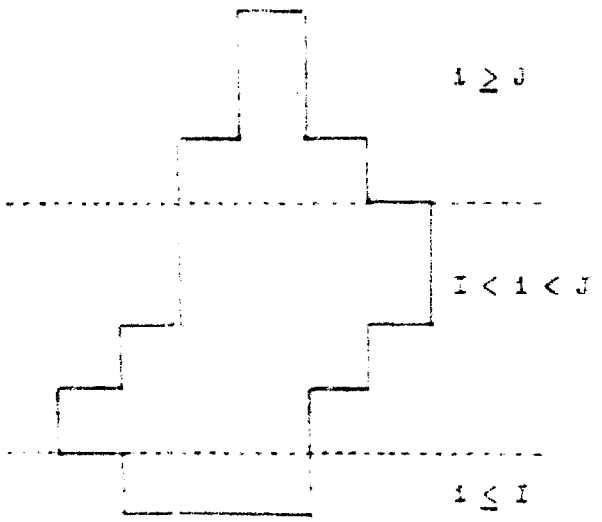

Fig. 1.

has radius of convergence 1 which exceeds r, it follows that $c(n) \sim 2c^*(n)$.

A trapezoid of base b_T and a parallelogram of base b_P can be joined in one way if $b_T = 0$ and $b_P - b_T$ ways otherwise. Hence $c^*(n)$ is the coefficient of $y_1^0 y_2^0$ in

$$G(x, y_1, y_2) = S(x, y_1)P(x, 1, y_1^{-1}, y_2^{-1})S(x, y_2)\,, \tag{8}$$

where

$$S(x, y) = T(x, y)\,\frac{y}{(1-y)^2} + \frac{1}{1-y}\,.$$

Thus

$$c(n) \sim -2r^{-n-1} \operatorname*{Res}_{x=r} C(y_1^0 y_2^0, G(x, y_1, y_2)) = fr^{-n}\,, \tag{9}$$

where $C(y^k, f(y))$ denotes the coefficient of y^k in the Laurent series for $f(y)$ and

$$\begin{aligned} f = (-2r^{-1} &\operatorname*{Res}_{x=r}(1 - P_2(x, 1, 1))^{-1}) \\ &\times C(y_2^0, S(r, y_2)P_1(r, 1, 1, y_2^{-1})) \\ &\times C(y_1^0, S(r, y_1)P_2(r, 1, y_1^{-1}))\,. \end{aligned} \tag{10}$$

The first factor in (10) is $2(rP_2', (r, 1, 1))^{-1}$. The second factor is (see [2, Theorem 1])

$$\frac{1}{2\pi i}\oint S(r, y)P_1(r, 1, 1, y^{-1})\,\frac{dy}{y} =$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1} r^{k(k+1)/2}}{2\pi i \prod_{j=1}^{n-1}(1-r^j)^2} \oint \frac{S(r, y)\,dy}{(y-r^k)y},$$

where the contour includes the origin, all singularities of P_1 and no singularities of S. The integrand has simple poles at $y = r^k$ and $y = 0$. Hence the second factor in (10) is

$$K = \sum_{k\geq 1} \frac{(-1)^{k+1} r^{k(k+1)/2}}{\prod_{j=1}^{n-1}(1-r^j)^2}\left(\frac{S(r, r^k)}{r^k} - \frac{S(r, 0)}{r^k}\right) = 1.02934.$$

Finally, we consider the last factor in (10). By rotating the plane 180°, we see that $P(x, 1, y, 1) = P(x, 1, 1, y)$. Multiplying this equation by $1 - P_2(x, 1, 1)$, substituting (6) into both sides, setting $x = r$, and using $P_2(r, 1, 1) = 1$, we obtain

$$P_2(r, 1, y)P_1(r, 1, 1, 1) = P_1(r, 1, 1, y).$$

Hence the third factor in (10) is $K/P_1(r, 1, 1, 1)$. Thus

$$f = 2K^2/(rP_1(r, 1, 1, 1)P_2'(r, 1, 1)) = 2.67483. \tag{11}$$

Comparing (11) and (7b), we see that about 22% of the convex animals are parallelograms. If we allow parallelograms which have been rotated 90°, then about 44% of the convex animals are parallelograms.

5. The shape of typical parallelograms

How do the height and thickness of n-cell parallelograms behave? We begin by studying the height. This information is contained in $P(z, w, 1, 1)$. By [1, Theorem 1], the probability that the height is less than h_n converges to a normal distribution with mean $\mu_n \sim 0.42088n$ and variance $\sigma_n^2 \sim 0.20806n$.

The width can be studied in various ways. From the previous paragraph we see that the average width is $n/\mu_n \sim 2.37597$. We can also study the base of a parallelogram by considering $P(x, 1, 1, y)$. Define

$$(12) \qquad \sum_n p_k(n)x^n = \left(y \frac{\partial}{\partial y}\right)^k P(x, 1, 1, y)\Big|_{y=1} .$$

It is easily seen that the base has mean and variance

$$\mu(n) = p_1(n)/p_0(n) \sim 2.06030,$$
$$\sigma^2(n) = p_2(n)/p_0(n) - \mu(n)^2 \sim (0.57609)^2 .$$

Since $\mu(n) < n/\mu_n$, parallelograms tend to be thicker near the middle.

We now show that if $f(n)/n \to \infty$, then the fraction of parallelograms whose slope is within $\pm f(n)$ of $\pm 45°$ goes to 1 as $n \to \infty$. Let ϵ and δ be small and positive. To begin with, "slope" is well defined to within $O(n^{\epsilon-1})$ since height is proportional to n and the bases are of bounded length, except for a fraction of parallelograms which goes to 0 as $n \to \infty$. Rotate a typical parallelogram by 90°. Except for a bounded number of cells near the bases, the result is a parallelogram. Both the old and new parallelograms have heights within $\delta f(n)$ of μ_n except for a fraction which goes to 0 as $n \to \infty$.

6. The shape of typical convex animals

The ideas in the previous section still apply, but we must work with equations like (8) instead of (6). All the results are the same except for the length of the bases to which the trapezoids are attached. We must replace (12) with

$$\sum_n c_k(n)x^n = C(y_1^0 y_2^0, S(x, y_1)P^{[k]}(x, 1, y_1^{-1}, y_2^{-1})S(x, y_2)) ,$$

where

$$P^{[k]}(x, 1, u, v) = \left(v \frac{v}{\partial v}\right)^k P(x, 1, u, v) .$$

Thus

$$c_k(n)/c_0(n) \sim K^{-1} C(y^0, P_1^{[k]}(r, 1, 1, y^{-1})S(r, y)) .$$

In particular

$$C(y^0, P_1^{[1]}(r, 1, 1, y^{-1})S(r, y)) =$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} r^{n(n+1)/2}}{2\pi i \prod_{k=1}^{n-1}(1-r^k)^2} \oint \frac{S(r, y)}{(y-r^n)^2}\, dy$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} r^{n(n+1)/2}}{\prod_{k=1}^{n-1}(1-r^k)^2} \left. \frac{\partial S(r, y)}{\partial y} \right|_{y=r^n} = 3.13352\,.$$

Since the average width is the same for parallelograms and convex animals while the end widths are different, it is likely that all widths except those near the top and bottom have a mean equal to the average width. I have not been able to prove this.

References

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